How to Install Node.js and Run Your First JavaScript Program on Windows

Step 1: Check if Node.js is Already Installed

Before installing Node.js, it's important to check if it's already installed on your Windows machine.

1. Open the **Command Prompt (CMD)**.
2. Type the following command:

   ```
   node -v
   ```

3. If you see the message `'node' is not recognized as an internal or external command...`, it means **Node.js** is not installed.

Step 2: Download Node.js

1. Open your **web browser**.
2. Type "download node js" in the search bar and press Enter.
3. Click on the link to **nodejs.org**.
4. On the Node.js website, you will see a download button for the latest version. There are two versions available:
   - **LTS** (Long-Term Support): Recommended for most users.
   - **Current**: For users who want the latest features.

   Note: Node.js is a JavaScript runtime environment. Do not confuse Node.js with JavaScript itself—**Node.js** is simply a runtime that allows you to run JavaScript outside the browser.

5. Click on the **LTS** download button to get the stable version.
6. Once the download is complete, run the installer.

Step 3: Install Node.js

1. **Run the Installer**: Open the downloaded installer.
2. **Set the Installation Path**: When prompted, choose the installation path (e.g., `C:\Program Files\nodejs`).
3. Follow the usual installation steps by clicking **Yes** and **Finish** when prompted.

Step 4: Verify Installation

After installation, it's important to verify that Node.js was installed correctly.

1. Open **Command Prompt (CMD)** again.
2. Type the following command to check the installed version of Node.js:

   ```
   node -v
   ```

It should display the version number of Node.js you just installed (e.g., `v16.x.x`).

3. Along with Node.js, the installer also installs **npm** (Node Package Manager). To verify that **npm** is installed, type:

   ```
   npm -v
   ```

   This will display the version of npm installed along with Node.js.

Why Do We Install Node.js and What Is npm?

What is Node.js?

- **Node.js** is a runtime that allows you to run **JavaScript** outside the browser. Since you're learning JavaScript for **Playwright**, you need **Node.js** to run Playwright scripts.
- It enables you to use JavaScript for automation tasks on the server-side, just like you use **Java** for Selenium tests.

In short: You install **Node.js** to run **JavaScript-based automation tools**, like **Playwright**, on your machine.

What is npm?

- **npm** (Node Package Manager) is used to manage **Node.js** libraries or dependencies, just like **Maven** in Java.
- When working with **Playwright** or any JavaScript testing tool, **npm** allows you to install the libraries you need.

In short: **npm** helps you install and manage packages like **Playwright** for running automation tests in **JavaScript**.

Using Playwright with Node.js

- To get started with **Playwright** in JavaScript, you need to:
  1. Install **Node.js**.
  2. Use **npm** to install **Playwright**.

     ```
     npm install playwright
     ```

  3. Write automation scripts in **JavaScript** using **Playwright** for browser automation.

Recap:

- **Node.js**: Needed to run **JavaScript** for tools like **Playwright**.
- **npm**: Manages JavaScript libraries (like **Playwright**), similar to **Maven** in **Java**.

Create and Run a Simple Hello World JavaScript Program

1. **Write the code**:
   - Open **Notepad** and write:

     ```
     console.log("Hello World");
     ```

2. **Save the file**:
   - Save it as **`MyFirstJavascriptProgram.js`** in the **Documents** folder.

3. **Navigate to the Documents folder in CMD**:
   - Open **Command Prompt (CMD)** and run:

     ```
     C:\Users\yogesh
     cd Documents
     ```

4. **Run the JavaScript file**:

   Execute the file with:

     ```
     node MyFirstJavascriptProgram.js
     ```

5. **Output**:
   - You should see:

     ```
     Hello World
     ```

Recap:

- **Write the program** in **Notepad**.
- **Save** it as **`.js`** in **Documents**.
- **Navigate to Documents** in **CMD** and run the program using **Node.js**.